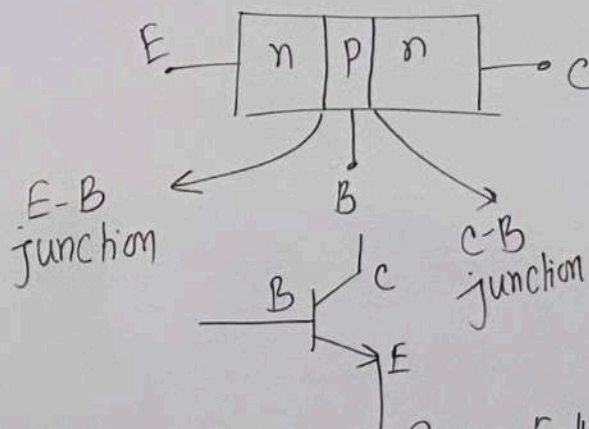
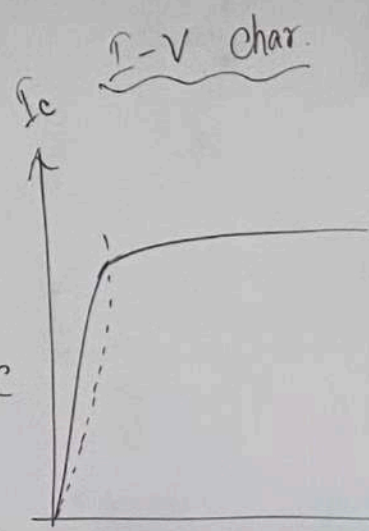


BJT

↳ npn
↳ pnp



Common Emitter Current gain $\rightarrow \beta$



$$I_E = I_B + I_C$$

cut-in voltage, $V_Y = 0.5V$

E-B junc.

C-B junc.

operating Modes

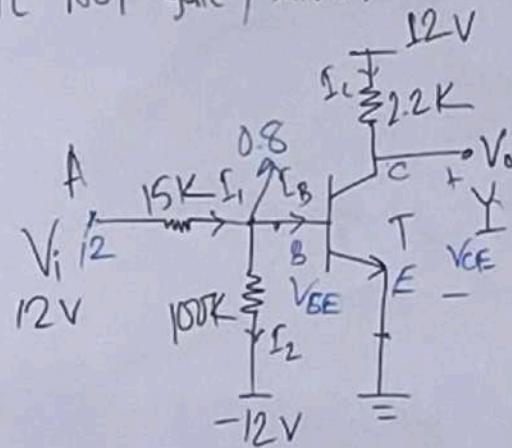
- i) FB
- ii) FB
- iii) RB
- iv) RB

- RB $\rightarrow$ Forward Active (on)
- FB $\rightarrow$ Saturation (on)
- RB $\rightarrow$ Cutoff (off)
- FB $\rightarrow$ Reverse Active

$V_{BE} > V_Y$, $I_C \neq 0, I_B \neq 0, I_E \neq 0$, $V_{BE} = 0.7V$, $\beta_F = \frac{I_C}{I_B} \rightarrow 30, 50, 100$
 $V_{BE} > V_Y$, $V_{BE} = 0.8V$, $V_{CE} = 0.2 \text{ or } 0.1V$, $\beta_{\text{forced}} = \frac{I_C}{I_B}$
 $V_{BE} < V_Y$, $I_C = I_B = I_E = 0$, $\beta_{\text{forced}} < 10$

RTL (Resistor Transistor Logic)

RTL NOT gate / Inverter



$$V_T = 0.5$$

$$\beta_F = 30$$

V_i	A	V_o	Y
0V	0	12	1
12V	1	0.2V	0

- Verify the above logic gate
- Find P , P_T , P_{avg}
- Operating Mode of T

Case-1

$V_i = 0V$, T $\rightarrow$ cutoff

$$I_c = I_b = I_E = 0$$

$$I_c = \frac{12 - V_o}{2.2K}$$

$$0 = 12 - V_o$$

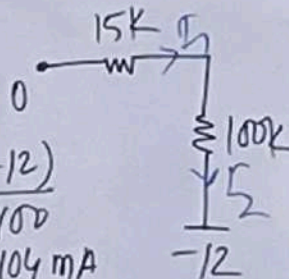
$$V_o = 12V$$

$$I_1 = I_2 = \frac{0 - (-12)}{15 + 100} = 0.104 \text{ mA}$$

$$P_1 = \Delta V \times I$$

$$= (0 - (-12)) \times 0.104$$

$$= 1.25 \text{ mW}$$



Case-2

$V_i = 12V$, T $\rightarrow$ Saturation

$$V_{BE} = 0.8V, V_{CE} = 0.2V$$

$$V_{CE} = V_c - V_E$$

$$0.2 = V_c - 0$$

$$V_c = 0.2V \rightarrow V_o = 0.2V$$

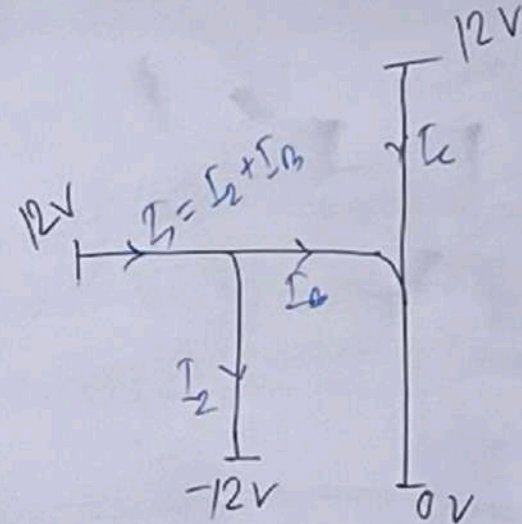
$$I_1 = \frac{12 - 0.8}{15} = 0.746 \text{ mA}$$

$$I_2 = \frac{0.8 - (-12)}{100} = 0.128 \text{ mA}$$

$$I_b = I_1 - I_2 = 0.61 \text{ mA}$$

$$I_c = \frac{12 - V_o}{2.2K} = \frac{12 - 0.2}{2.2} = 5.36 \text{ mA}$$

$$I_E = I_b + I_c = 5.97 \text{ mA}$$



$$P_2 = (12 - (-12)) \times I_2 + (12 - 0) \times I_b$$

$$+ (12 - 0) \times I_c$$

$$= 74.808 \text{ mW}$$

$$P_T = P_1 + P_2$$

$$P_{avg} = \frac{P_1 + P_2}{2}$$

$$\beta = \frac{I_c}{I_b}$$

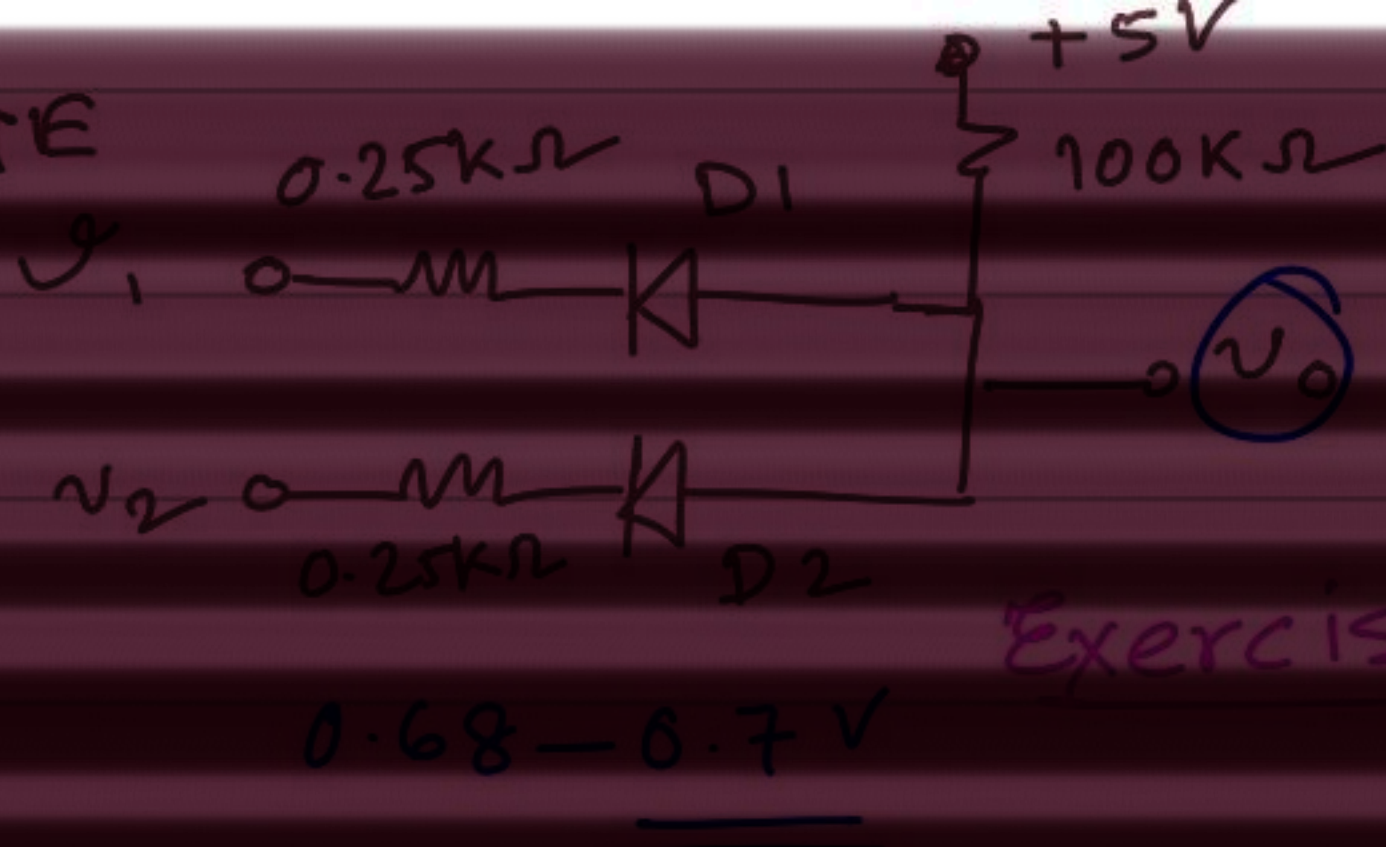
$$\Rightarrow \beta = \frac{5.36}{0.61}$$

$$\beta = 8.67 < 10$$

$\therefore$ Saturation

AND GATE: 2-AND GATE

v_1	v_2	v_o	D1	D2
5V	5V	5V	OFF	OFF
5V	0V	✓	OFF	ON
0V	5V	✓	ON	OFF
0V	0V	✗	ON	ON

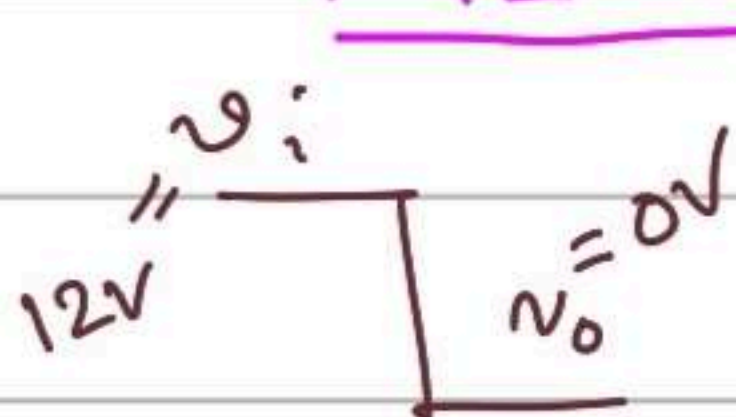


For choosing output logical low, we will take max of the logical output low.

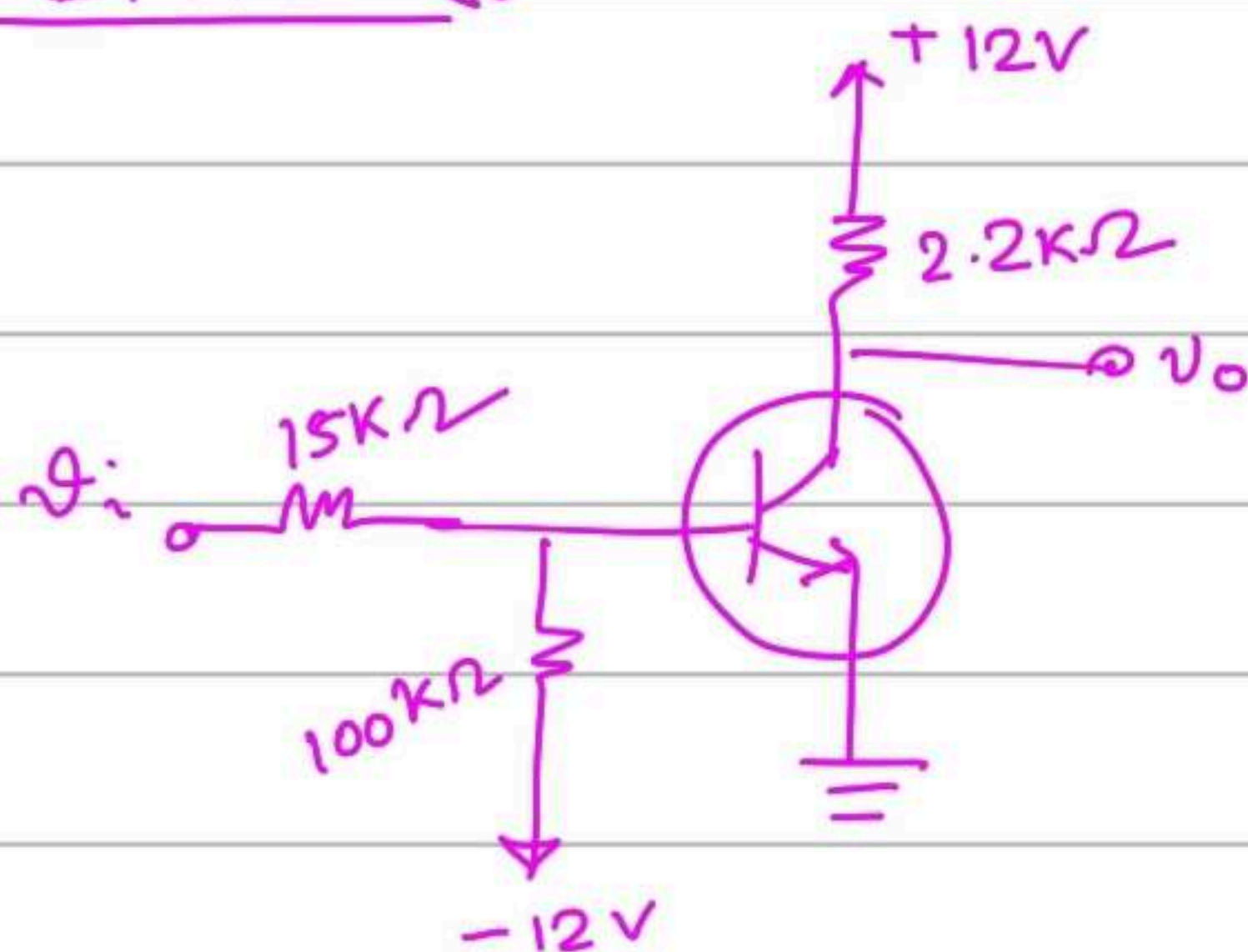
$V_{OL} = 0.7V$ $\rightarrow$ all voltages below this voltage would be considered logical low.

Register Transistor Logic (RTL)

RTL INVERTER:-



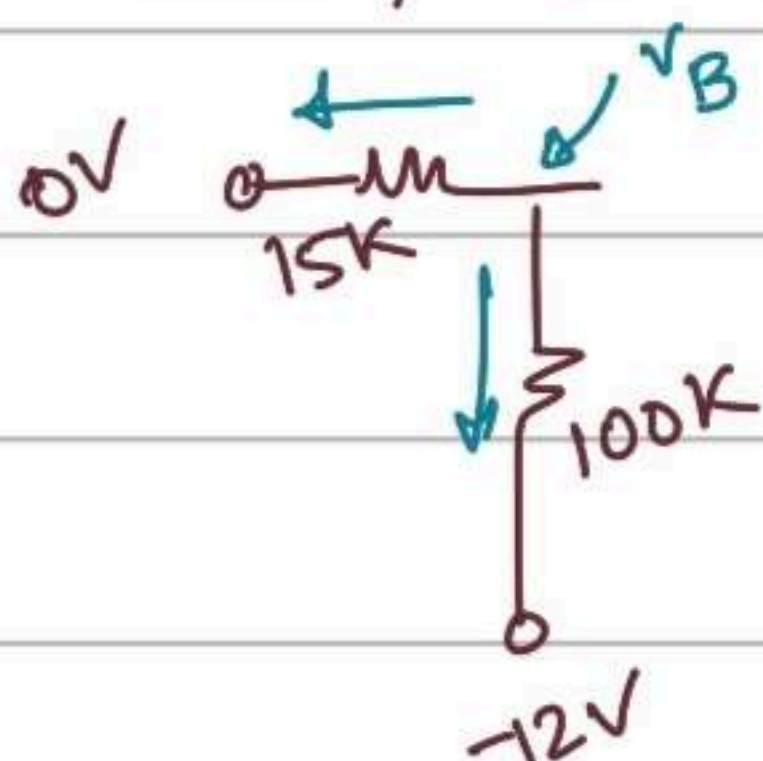
input logical
 $v_i = +12V$
low = 0V



v_i	v_o
1	0
0	1

Case 1: $v_i = 0V$

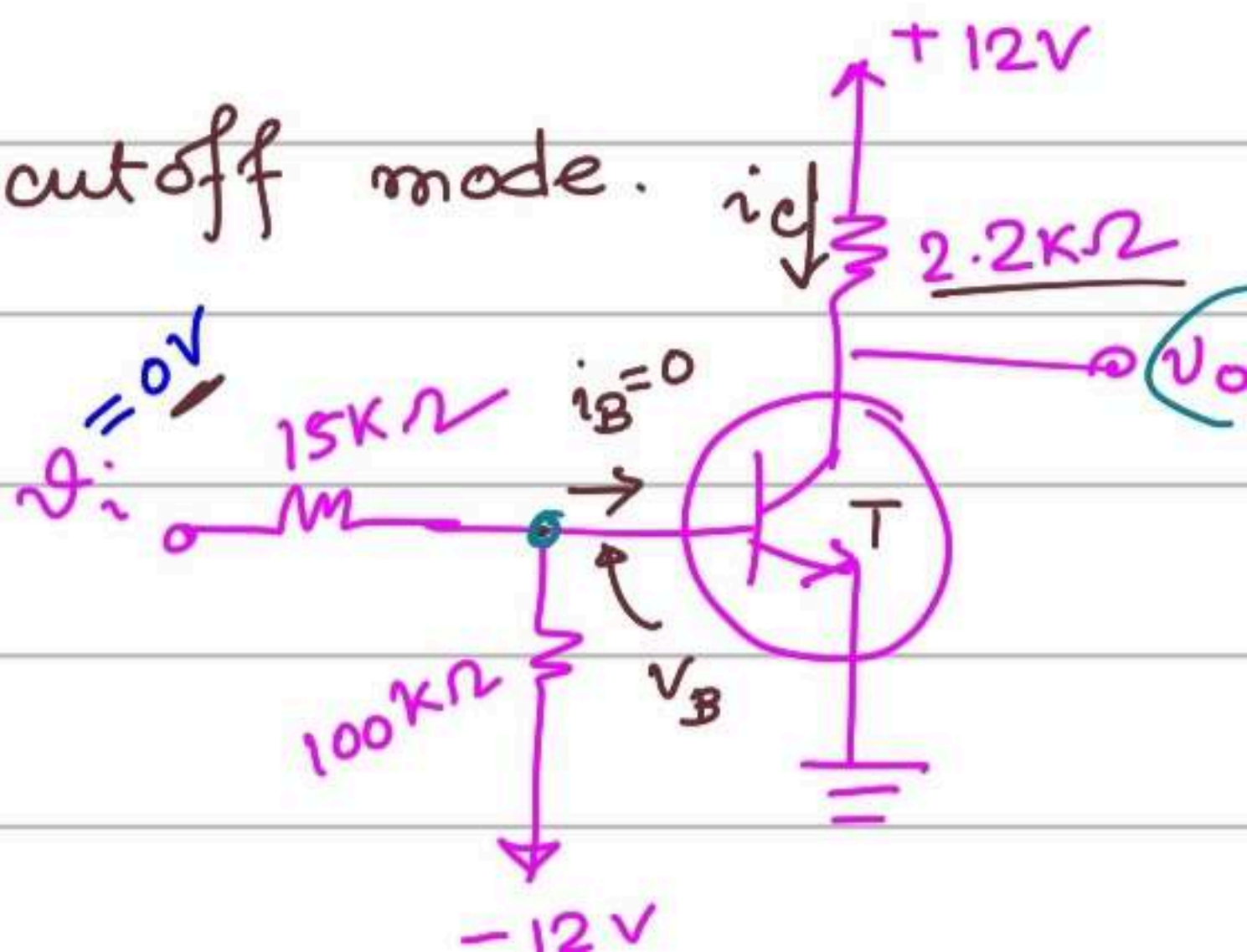
Guess/Assumption: T is in cutoff mode.



$$\frac{v_B - (-12)}{100k} + \frac{v_B - 0}{15k} = 0$$

$$\Rightarrow v_B = -1.565V$$

$$v_E = 0V$$



Verification: $V_{\gamma} = V_{BE}$ (cutin voltage) $\leq 0.5V$

$$V_{BE} = -1.565V < 0.5V \quad (\text{Our assumption is valid})$$

For cutoff mode. $i_C = i_B = i_E = 0mA$

$$i_C = \frac{12 - v_o}{2.2K} = 0 \Rightarrow \boxed{v_o = 12V} \quad \text{logical high output voltage}$$

Case ②: $v_i = +12V$

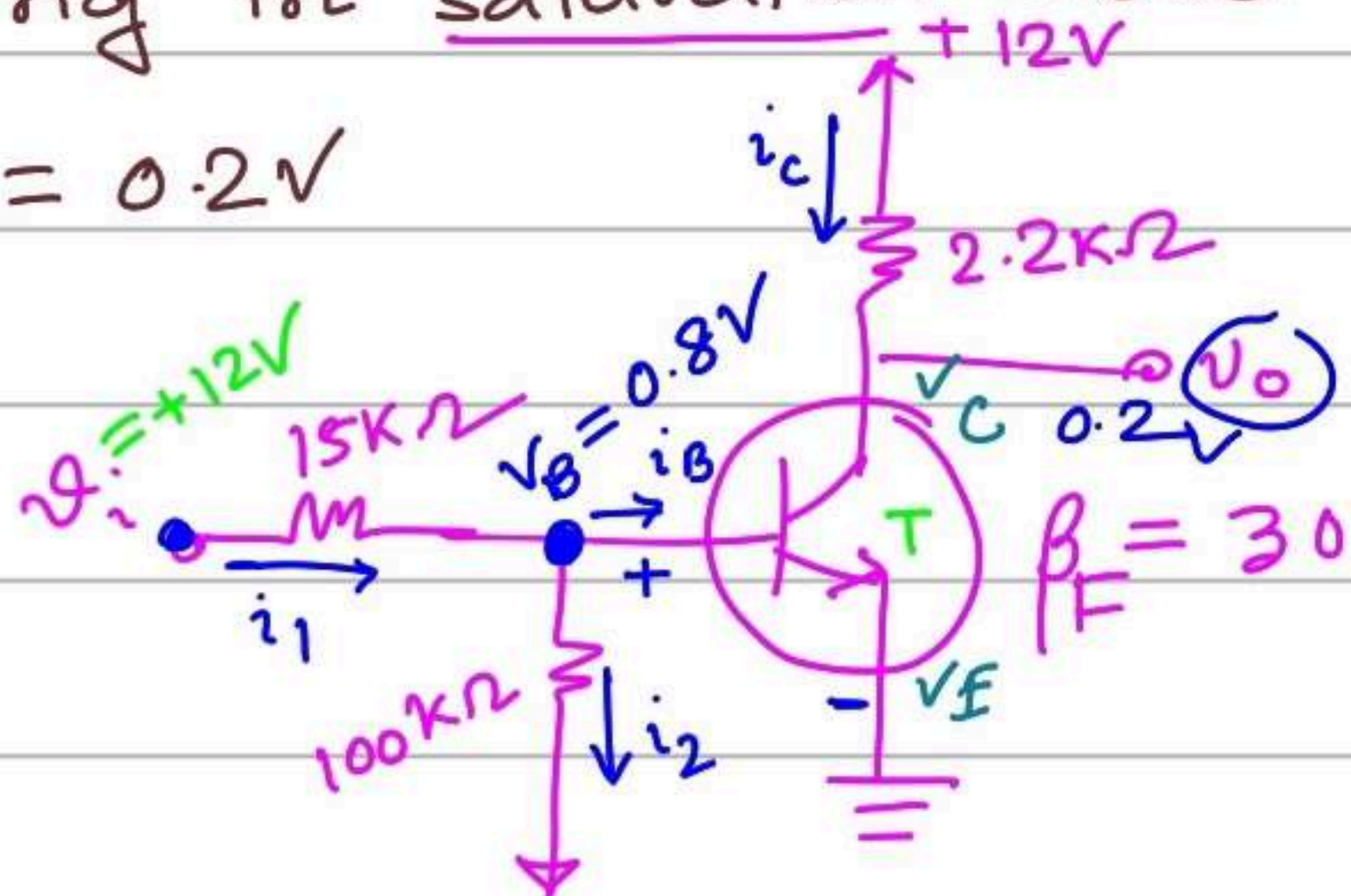
Assumption: T is operating in saturation mode.

$$\star v_o = v_C = v_C - v_E = v_{CE} = 0.2V$$

$$\star i_C = \frac{12 - 0.2}{2.2K} = 5.36mA$$

$$\star i_1 = \frac{12 - 0.8}{15K} = 0.746mA$$

$$\star i_2 = \frac{0.8 - (-12)}{100K} = 0.128mA$$



When transistor is operating in saturation mode $V_{BE} = 0.8V$, $V_{CE} = 0.2V/0.1V$

$$i_B = i_1 - i_2 = 0.619mA$$

Verification: $\beta_{\text{forced}} = \frac{i_C}{i_B} = 8.659$

Our assumption is valid.

$$< 30 = \beta_F$$

Thus our analysis shows output voltage

is now $0.2V$. = logical low voltage at the output terminal.
 $\boxed{V_{OL} = 0.2V}$

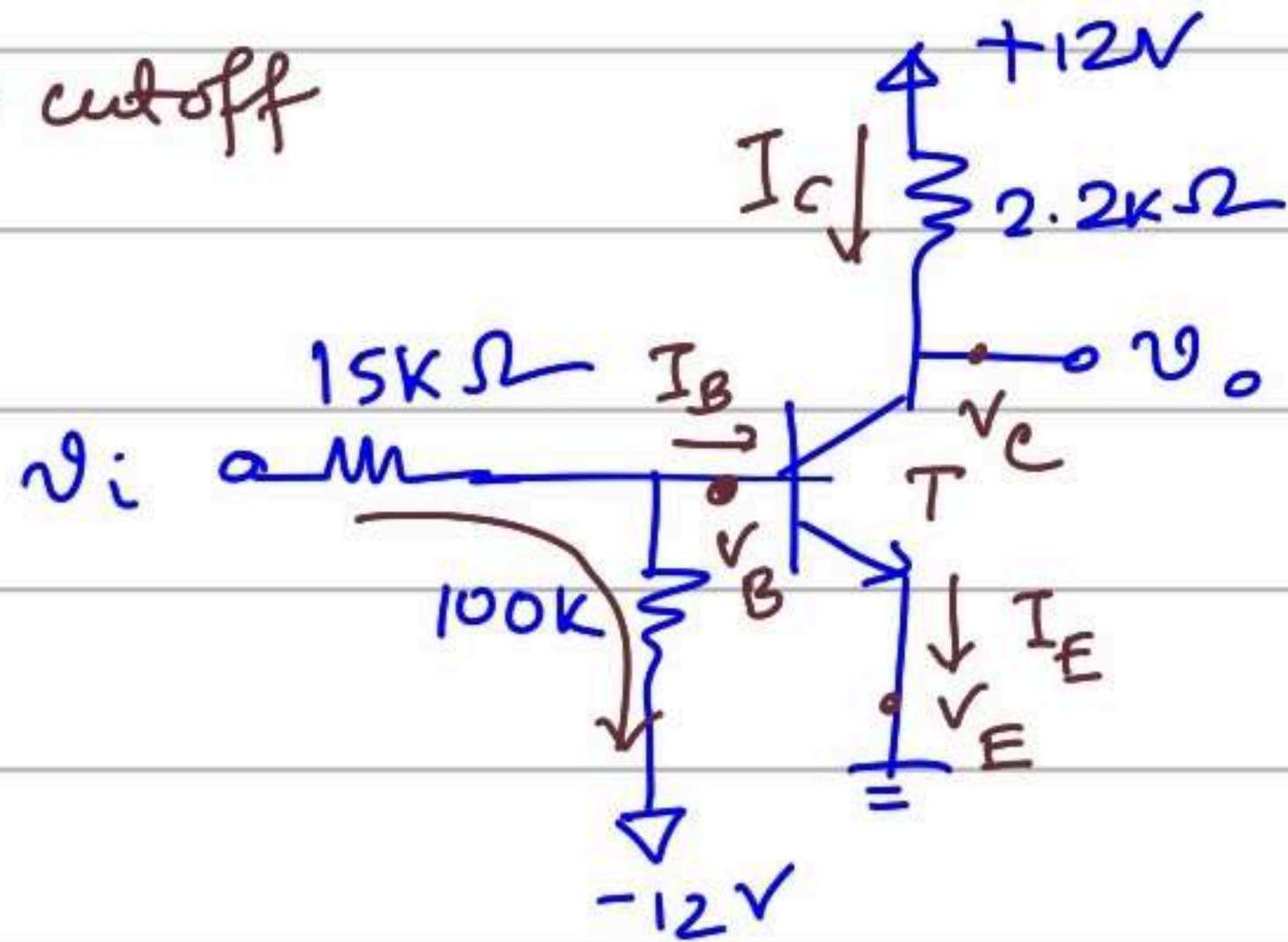
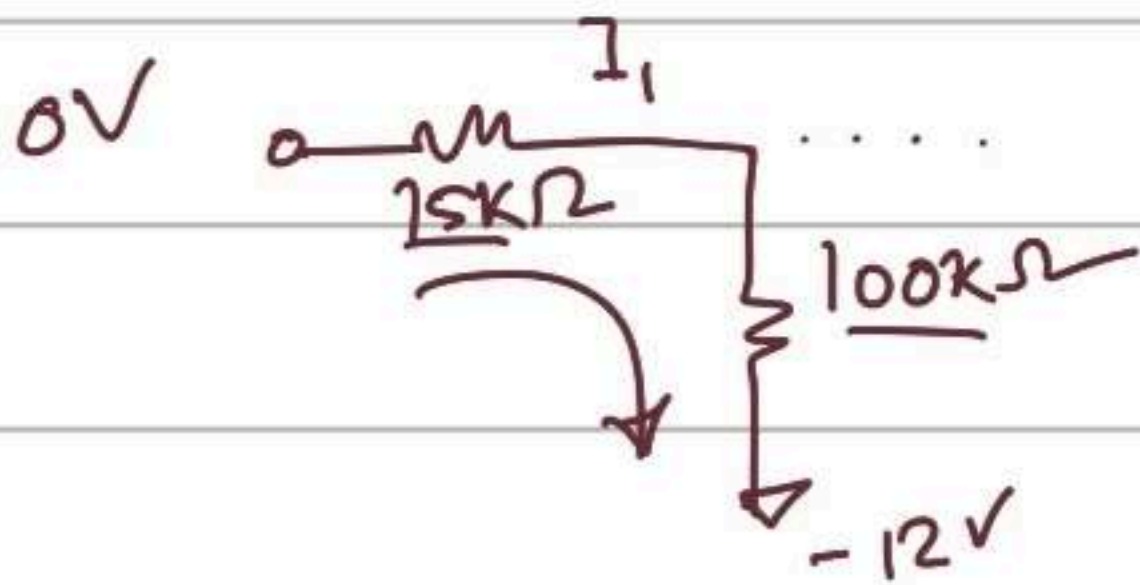
RTL circuits : Power dissipation,

Noise Margin and Fanout Calculation.

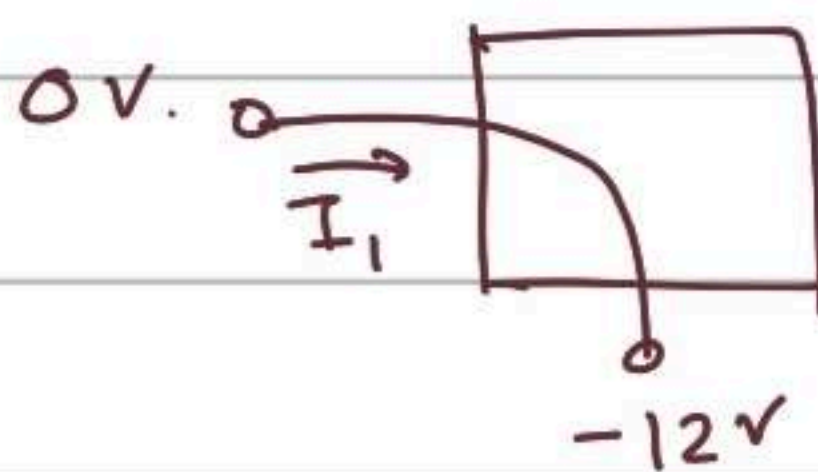
Power dissipation in RTL inverter :-

Case ①. $v_i = 0V$. $T \rightarrow$ cutoff

$$I_B = I_C = I_E = 0mA$$



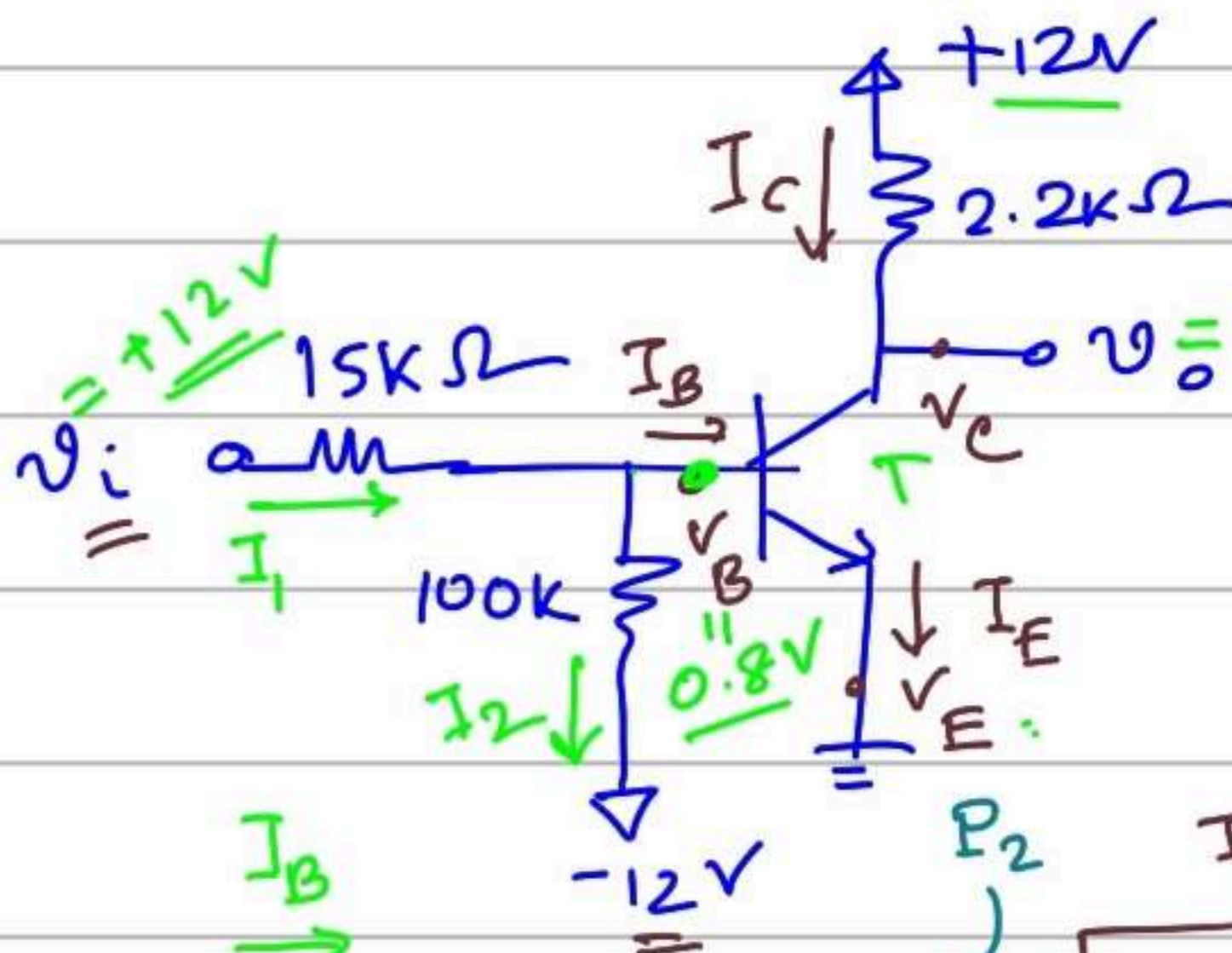
$$I_1 = \frac{0 - (-12)}{115k\Omega} = 0.104mA$$



$$P = (0 - (-12)) \times 0.104 = \boxed{1.2521mW}$$

Case ② $v_i = +12V$. $T \rightarrow$ saturation (RTL)

$$V_{BE} = 0.8V, V_{CE} = 0.2V \text{ (0.1V)}$$

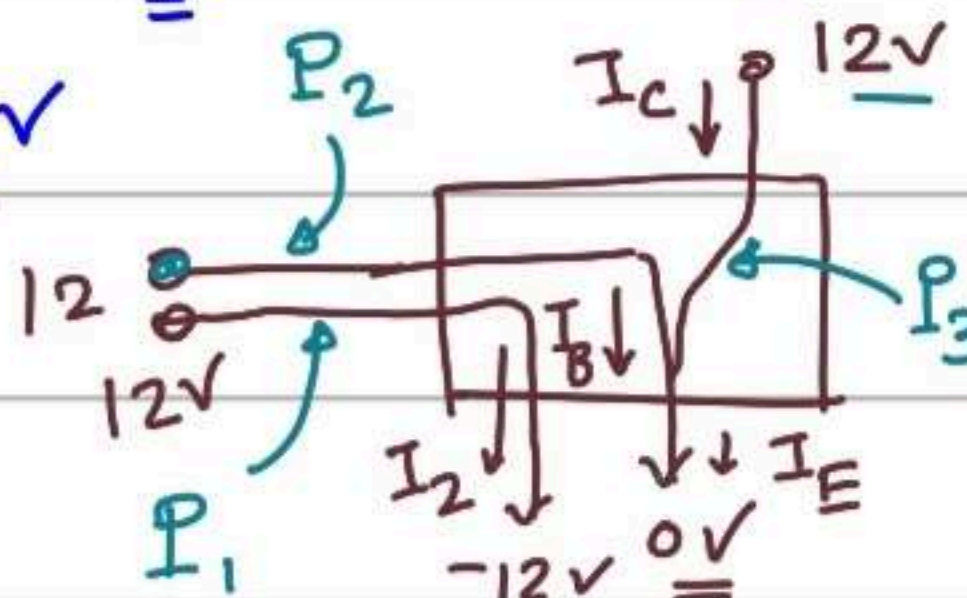
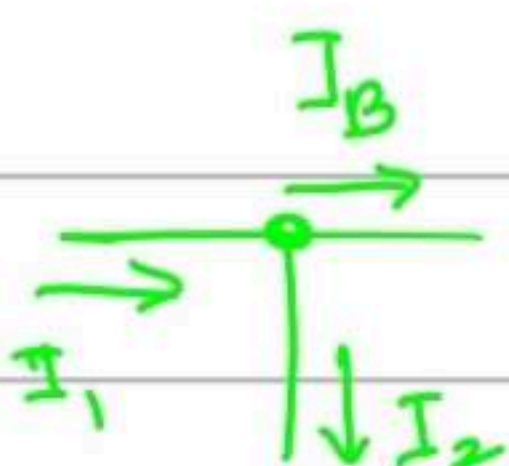


$$I_C = \frac{12 - 0.2}{2.2k} = 5.3636mA$$

$$I_1 = \frac{12 - 0.8}{15k} = 0.7466mA$$

$$I_2 = \frac{0.8 - (-12)}{100k} = 0.128mA$$

$$I_B = I_1 - I_2 = 0.6186mA$$

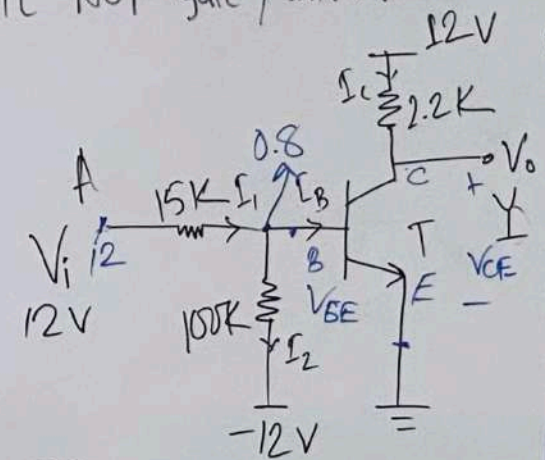


$$P_1 = (12 - (-12)) \times 0.128 = 3.072mW$$

$$P_2 = (12 - 0) \times 0.6186 = 7.4323mW$$

#RTL (Resistor Transistor Logic)

RTL NOT gate / Inverter



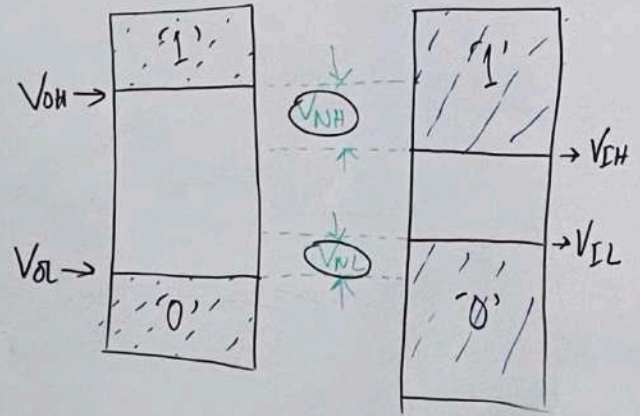
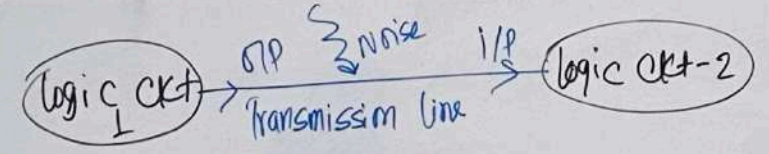
$V_{\gamma} = 0.5$
 $B_f = 30$

V_i	A	V_o	Y
0V	0	12	1
12V	1	0.2V	0

(iv) Find the Noise Margin

Noise Margin

The amount of Noise Voltage a logic ckt can tolerate while successful transmission.



$V_{NH} \rightarrow$ High state Noise Margin

$$V_{NH} = V_{0H} - V_{1H}$$

$V_{NL} \rightarrow$ Low state Noise Margin

$$V_{NL} = V_{1L} - V_{0L}$$

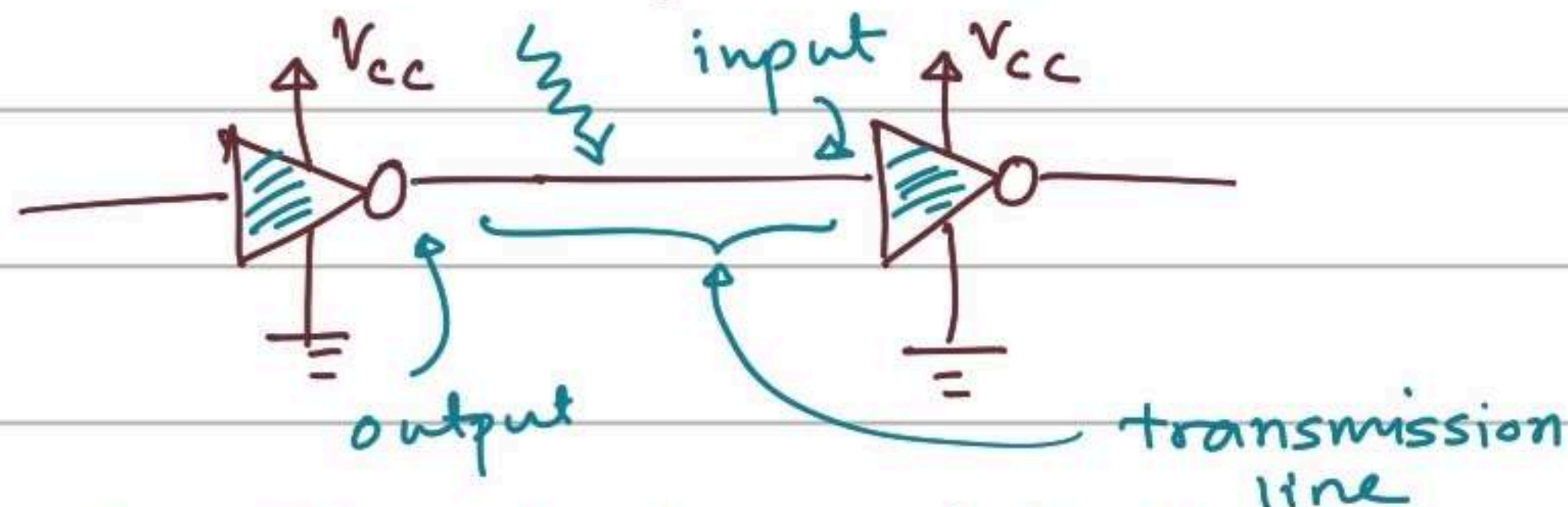
Max/overall Noise Margin $\rightarrow V_{NM} = \min(V_{NL}, V_{NH})$

$$P_3 = (12 - 0) \times 5.3636 = 64.3632 \text{ mW}$$

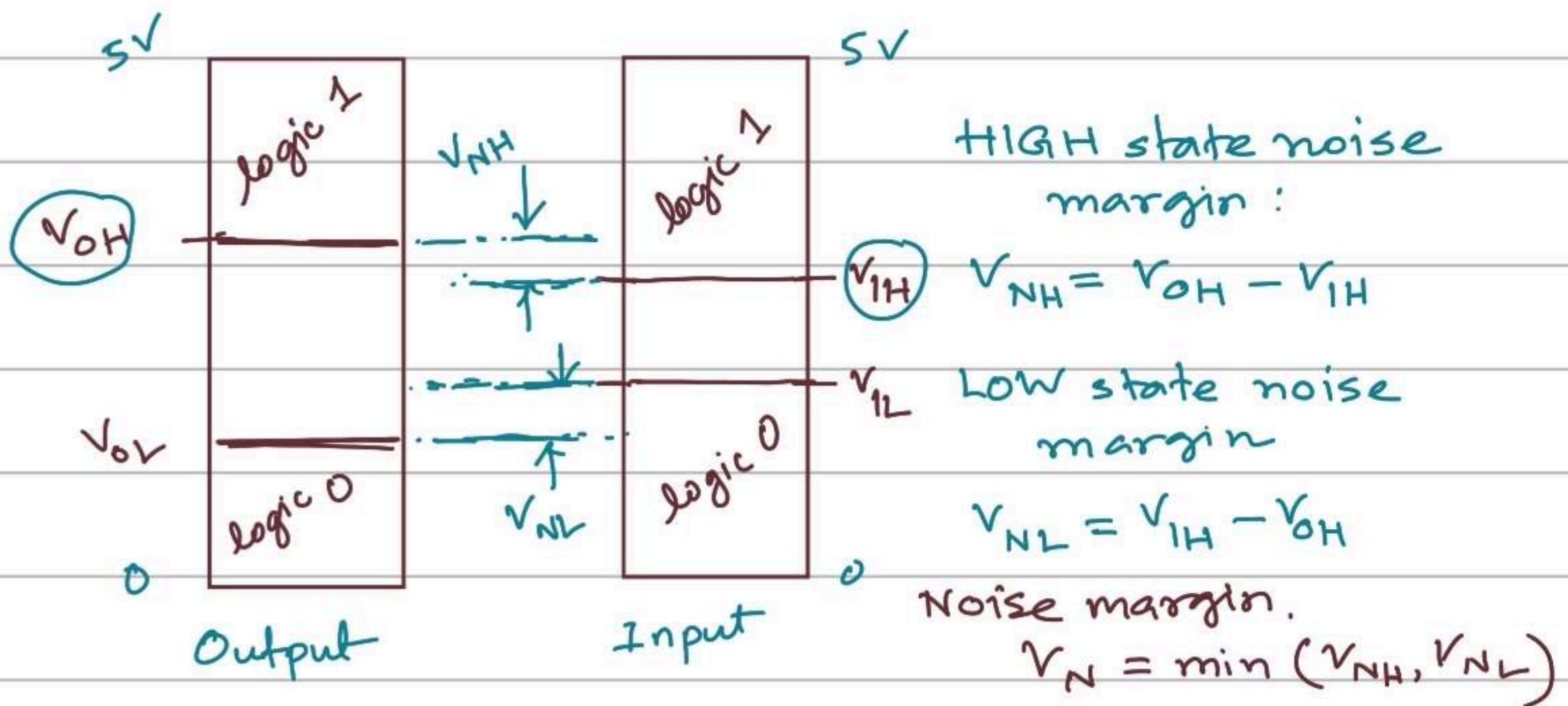
$$\text{Total power dissipation, } P = P_1 + P_2 + P_3 = \boxed{74.8534 \text{ mW}}$$

NOISE MARGIN:-

The amount of noise voltage a ^{digital logic} circuit can tolerate while successful transmission.



We model different types interference or noises using "noise voltage."



Why do we use minimum?

→ To ensure safety margins for both high and low voltages.

$\frac{V_{OH}}{V_{IH}}$ → minimum voltage level at an output in the logical "1" state under defined load condition.

$V_{OL} \rightarrow$ maximum " " at an output ^{input} in the logical "0" state under defined load condition.

Example: Output voltage is low when T is in saturation. $V_{OL} = 0.2V$ (Saturated logic family)

* $V_{OH} = ?$ (This will be given)

The output ^{high} voltage can be allowed to decrease 0.5V from its maximum value.

* $V_{OH} = +12 - 0.5V = 11.5V$

* V_{IL} calculation:

* When input voltage is low, transistor operates in cutoff mode.

* For RTL circuit, V_{IL} should be equal to the maximum voltage that takes the transistor on the verge of turn off to turn on condition.

* The transistor will turn on if $V_{BE} \geq 0.5V$.

* We need find the input voltage so that

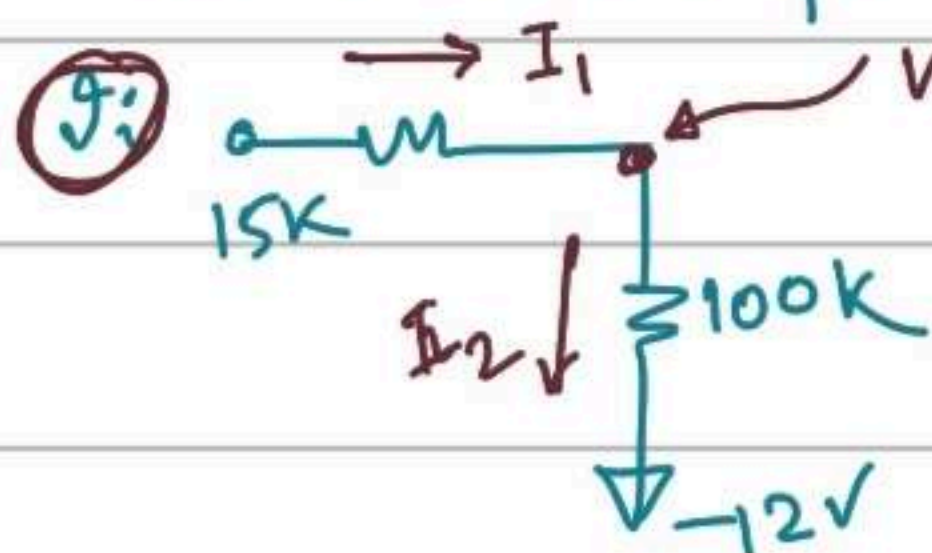
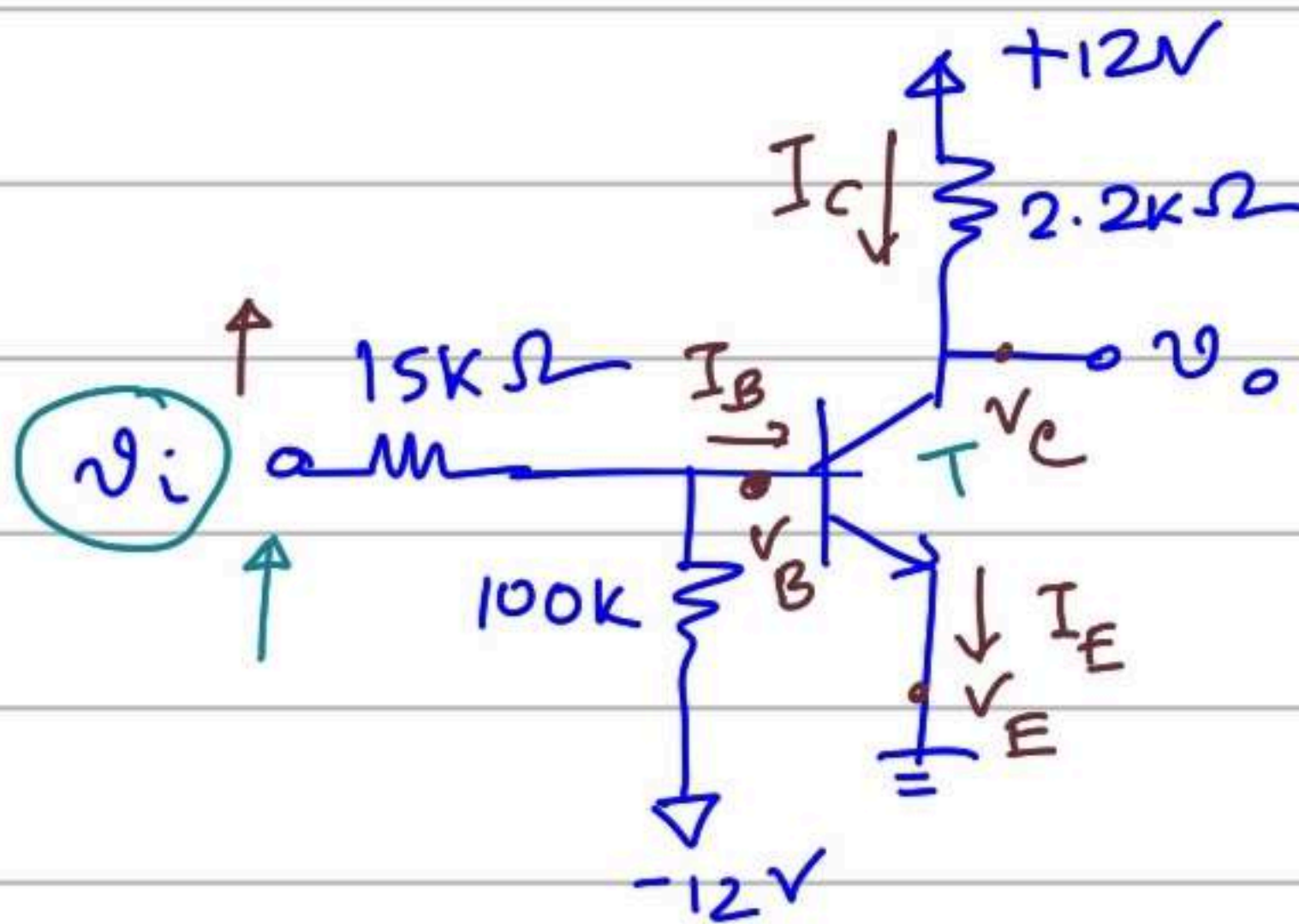
$V_{BE} = 0.5V$ but we assume transistor is in cutoff.

* T $\rightarrow$ cutoff, $I_B = 0mA$ $I_2 = \frac{0.5 - (-12)}{100k} = 0.125mA$

$I_1 = I_2$, $I_1 = \frac{V_i - 0.5}{15k} = 0.125mA$

$\therefore V_i = 0.5 + 15 \times 0.125 = 2.375V$

$V_{IL} = 2.375V$



V_{IH} calculation: When input voltage is high, transistor

T operates in saturation mode. $V_{CE} = 0.2V = v_o$.

* The minimum voltage at the input terminal which will change the operating mode of transistor

T to saturation to forward active mode should be V_{IH} .

* We will assume T is saturation,

but $\beta_{forced} \approx \beta_F$. [Transition point between sat. and F.A.]

* T $\rightarrow$ saturation. $I_C = \frac{12 - 0.2}{2.2k} = 5.3636 \text{ mA}$

$$\beta_{forced} = \frac{I_C}{I_B} = 30 \Rightarrow I_B = \frac{I_C}{\beta} = 0.1788 \text{ mA}.$$

$$I_2 = \frac{0.8 - (-12)}{100k} = 0.128 \text{ mA}. \quad I_1 = I_2 + I_B = 0.3068 \text{ mA}$$

$$\frac{v_i - 0.8}{15k} = 0.3068 \text{ mA} \Rightarrow v_i = 0.8 + 15 \times 0.3068 = 5.4018 \text{ V}$$

$$\boxed{V_{IH} = 5.4018 \text{ V}}$$

$$\text{High state noise margin: } V_{NH} = V_{OH} - V_{IH} = 6.0982 \text{ V}$$

$$\text{Low state noise margin: } V_{NL} = V_{IL} - V_{OL} = 2.175 \text{ V}$$

$$\text{Noise margin} = \min(6.0982, 2.175) = \boxed{2.175 \text{ V}}$$

